Lecture 7: Bayesian Analysis

Math for Data Science · 2026

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Agenda

- Introduction to Bayesian Data Analysis
- Part I: Black Box the Bayesian Way
- Part II: Summarising the Posterior
- Part III: Prediction
- Part IV: The Kidney Cancer Example

Introduction to Bayesian Data Analysis

The Process of Bayesian Data Analysis

1. Setting up a full probability model
    - a joint probability distribution for all observable and unobservable quantities
    - consistent with knowledge about the underlying scientific problem and the data collection process
2. Conditioning on observed data
    - calculating and interpreting the appropriate posterior distribution - the conditional probability distribution of the unobserved quantities of ultimate interest, given the observed data
3. Evaluating the fit of the model
    - are the conclusions reasonable?
    - how sensitive is the model to assumptions?

Conditioning in Bayesian Inference

- Bayesian conclusions about parameters or unobserved data are expressed as probability statements.

$$p(\theta \mid x), \text{ where } p(\theta) \text{ is, e.g., } \mathcal{N}(\theta \mid \mu, \sigma^2)$$

- These are conditional on the observed data x.
- The key distinction from classical inference:
  - Bayesian inference conditions on observed x.
  - Frequentist inference evaluates estimators over hypothetical samples of x given true θ.
- Thus, Bayesian analysis treats data as fixed and parameters as random, while frequentist analysis does the reverse.

Bayes' Rule and the Posterior Distribution

Bayes' rule:

$$p(\theta \mid x) = \frac{p(\theta)\, p(x \mid \theta)}{p(x)}$$

Marginal likelihood, or evidence:

$$p(x) = \sum_{\theta} p(\theta)\, p(x \mid \theta) \quad \text{or} \quad p(x) = \int p(\theta)\, p(x \mid \theta)\, d\theta$$

Unnormalized posterior:

$$p(\theta \mid x) \propto p(\theta)\, p(x \mid \theta)$$

Interpretation:

- $p(x \mid \theta)$: likelihood (function of θ)
- $p(\theta)$: prior
- $p(x)$: evidence (a constant, not a function of θ)
- $p(\theta \mid x)$: posterior: combines prior and likelihood

Part I: Black Box the Bayesian Way

The Black Box, Read the Bayesian Way

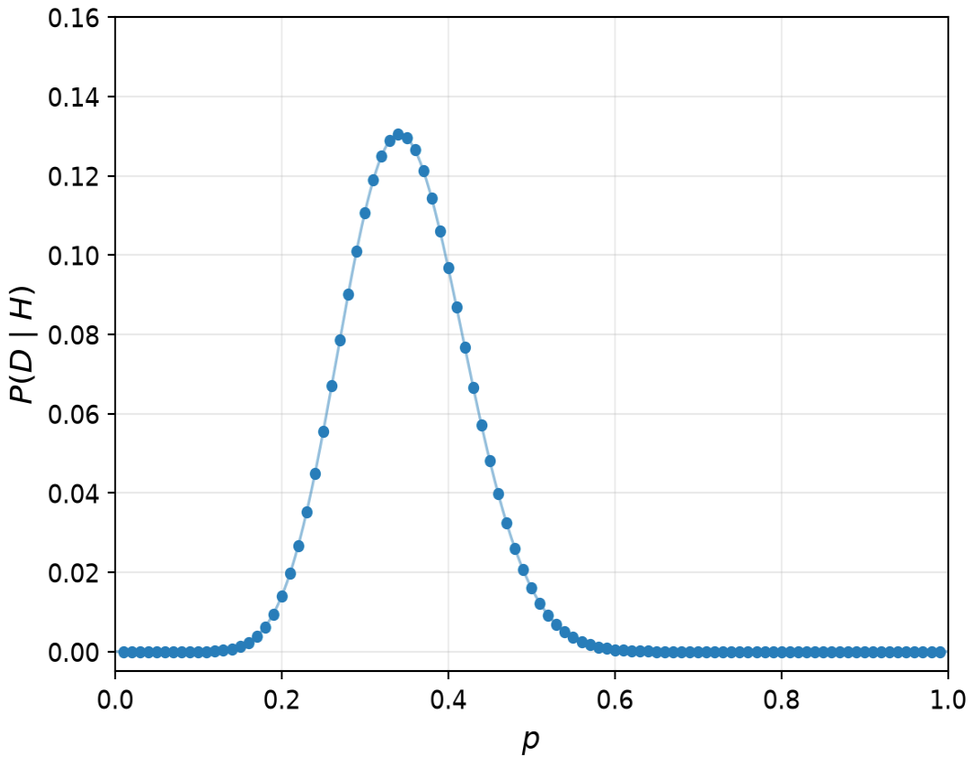
- Recall the likelihood of the black box, 14 wins in 41 tries:

$$\mathcal{L}_n(\theta) = p(x \mid \theta) = \binom{41}{14} \theta^{14}(1-\theta)^{27}$$

- We got there by asking how probable the observed data is *conditional on a hypothesis* for the parameter. We called it $p = P(🟡🟡)$; now it is θ.

- The likelihood is not a probability distribution over θ: it does not integrate to one. We normalised it, and that led us to the Beta distribution, $\text{Beta}(15, 28)$.

- The result was a probability density for the parameter. That step was already Bayesian, with a uniform prior $p(\theta) = 1$: the normalised likelihood *is* the posterior, $\theta \mid x \sim \text{Beta}(15, 28)$.



The Beta Distribution

Beta distribution with the Beta function $B(\alpha, \beta)$:

$$p(\theta \mid \alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \qquad \theta \in [0, 1]$$

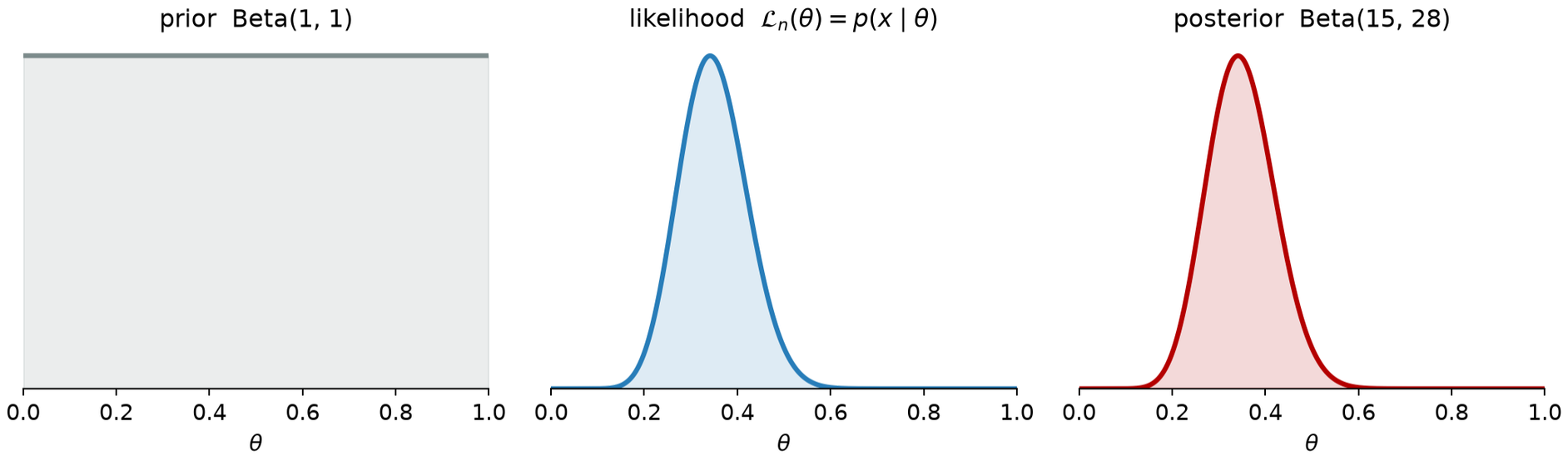
The Beta function can be written with the Gamma function Γ:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)} \qquad \Rightarrow \qquad p(\theta \mid \alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\,\Gamma(\beta)}\,\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

For a positive integer n the Gamma function is a factorial, $\Gamma(n) = (n-1)!$.

The Gamma function is the extension of the factorial to complex numbers.

The Flat Prior, Made Explicit



The flat prior is itself a Beta: $p(\theta) = 1$ is Beta$(1, 1)$.

With a flat prior, the evidence is the area under the likelihood:

$$p(x) = \int_0^1 p(\theta)\, p(x \mid \theta)\, d\theta = \binom{41}{14} \int_0^1 \theta^{14}(1-\theta)^{27}\, d\theta$$
$$= \binom{41}{14} \frac{\Gamma(15)\,\Gamma(28)}{\Gamma(43)} = \frac{41!}{14!\,27!} \cdot \frac{14!\,27!}{42!} = \frac{1}{42}$$

With Constant Prior The Peak Is the MLE

With $p(\theta) = 1$ the posterior and the likelihood differ only by a constant, so they peak at the same place:

$$\arg\max_\theta \; p(\theta \mid x) = \arg\max_\theta \; \mathcal{L}_n(\theta) = \frac{14}{41} \approx 0.34$$

The peak of $\mathcal{L}_n(\theta)$ is the maximum likelihood estimate $\hat{\theta}$.

The posterior keeps the whole curve: how plausible every other θ still is.

What If We Thought that the box is fair?

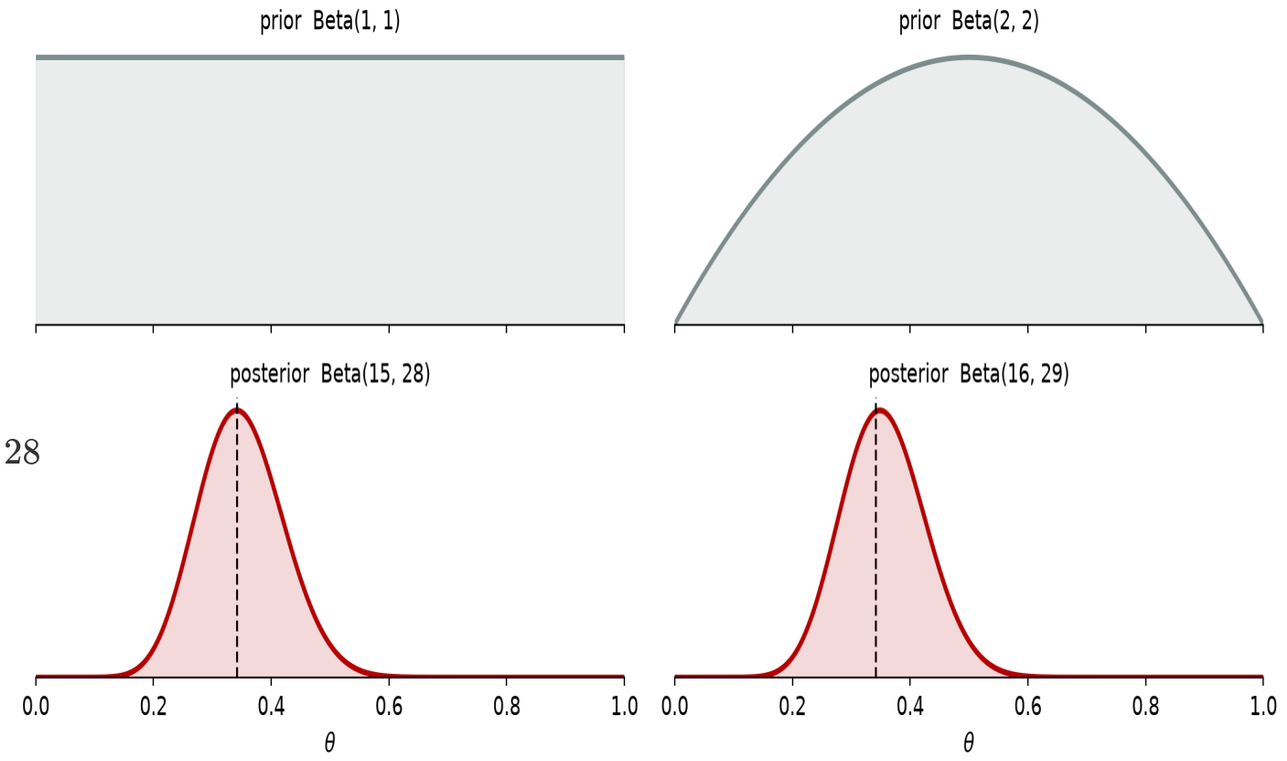
Suppose we believe, before any try, that the box is fair. A Beta$(2, 2)$ prior gives the center more probability.

$$p(\theta) \propto \theta\,(1-\theta)$$

Multiply by the likelihood and add the exponents:

$$p(\theta \mid x) \propto \underbrace{\theta\,(1-\theta)}_{\text{prior}} \cdot \underbrace{\theta^{14}(1-\theta)^{27}}_{\text{likelihood}} = \theta^{15}(1-\theta)^{28}$$

The posterior is Beta$(16, 29)$. It moves only slightly towards $\frac{1}{2}$.



Multiply and Collect

That worked because the prior and the likelihood have the same shape. The same step for any Beta(α, β) prior and any k wins in n tries:

$$p(\theta \mid x) \;\propto\; \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}}_{\text{prior}} \cdot \underbrace{\theta^{k}(1-\theta)^{n-k}}_{\text{likelihood}}$$

Both factors are powers of θ and $1-\theta$, so we just add exponents:

$$p(\theta \mid x) \;\propto\; \theta^{\,(\alpha+k)-1}\,(1-\theta)^{\,(\beta+n-k)-1}$$

That is the shape of a Beta again. And since there is only one density with that shape, we can read the answer straight off:

$$\theta \mid x \;\sim\; \mathrm{Beta}(\alpha+k,\; \beta+n-k)$$

No integral for the evidence this time. We recognised the shape, and normalisation took care of itself.

Conjugacy

Definition. A prior is conjugate to a likelihood when the posterior belongs to the same family as the prior.

The Beta is conjugate to the Binomial. Updating is then pure bookkeeping:

$$(\alpha, \beta) \longmapsto (\alpha + k,\ \beta + n - k)$$

Add wins to α, losses to β. So the parameters act as pseudo-counts: compared with the flat Beta$(1, 1)$, a Beta(α, β) prior has already seen $\alpha - 1$ wins and $\beta - 1$ losses before the experiment began.

What If We Thought that the box is rigged?

Suppose we believe, before any try, that the box is rigged. It looks like a gambling machine and we are suspicious.

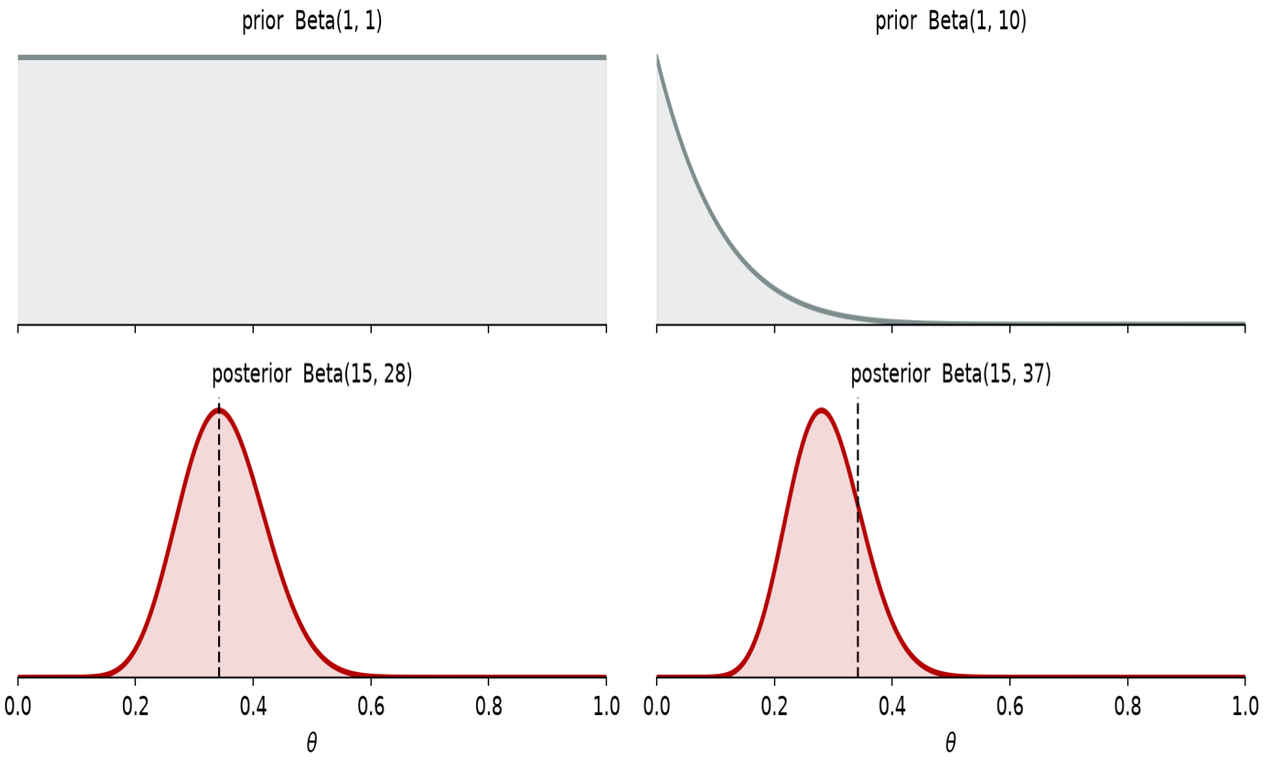
A Beta$(1, 10)$ prior gives the left side more probability: nine pretend losses and no pretend wins.

$$p(\theta) \propto (1-\theta)^9$$

No derivation this time. Add the 14 wins and the 27 losses:

$$(1, 10) \;\longmapsto\; (1+14,\; 10+27) = (15, 37)$$

The posterior is Beta$(15, 37)$. The peak drops from 0.34 to 0.28.

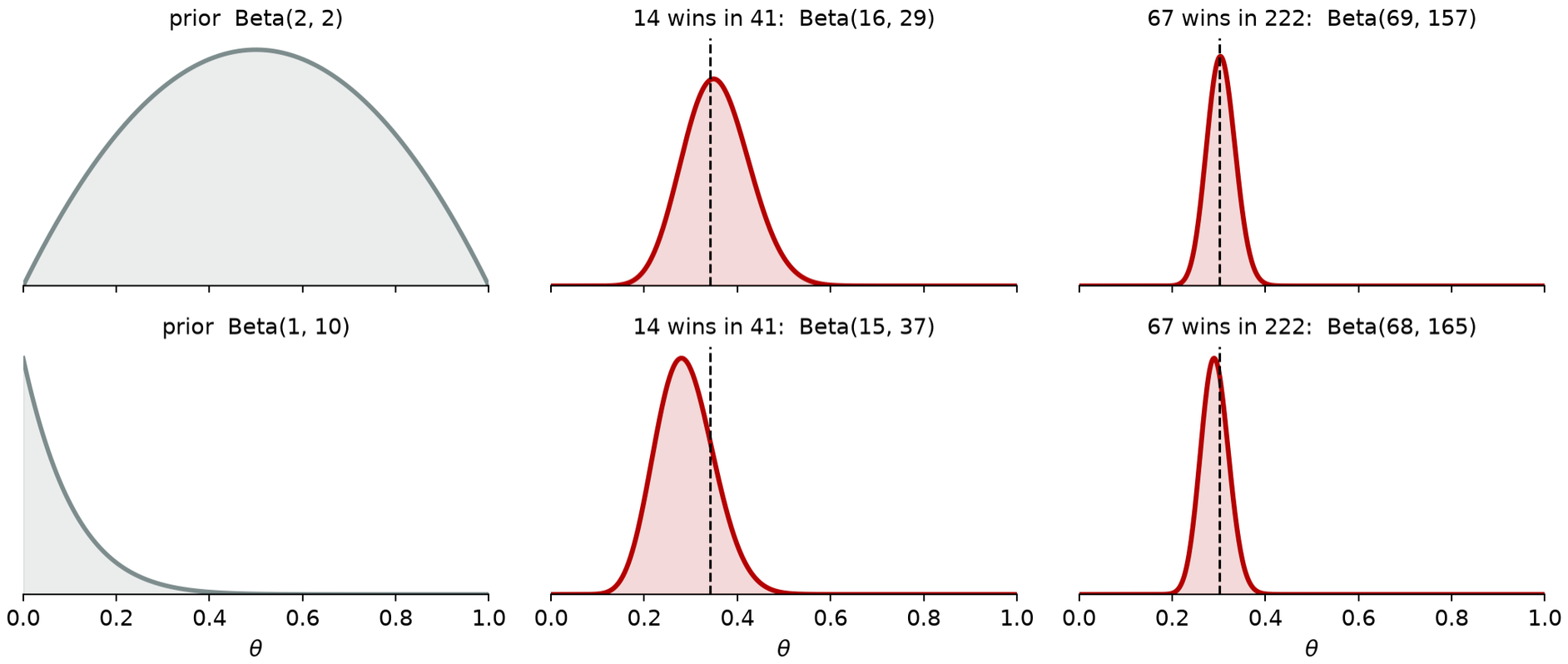


Impact of the Prior

Prior	Pretend data beyond the flat prior	Says
Beta$(1, 1)$	none	every θ equally credible
Beta$(2, 2)$	1 win, 1 loss	mild pull to the middle, extremes unlikely
Beta$(1, 10)$	9 losses	“this machine rarely pays out”

The honest question is never *“is this prior true?”* but *“how many observations is it worth, and can I defend that?”*

Posterior as a Compromise Between Data and Prior Information



After 41 tries the two priors still disagree: the posteriors peak at 0.35 and 0.28.

After another 53 wins in 181 tries, 67 wins in 222 altogether, they peak at 0.30 and 0.29: the prior has washed out.

Yesterday's Posterior Is Today's Prior

Because updating only adds to the pseudo-counts, we can feed the data in any order, in any batches:

$$(\alpha, \beta) \xrightarrow{n_1, k_1} (\alpha + k_1,\ \beta + n_1 - k_1) \xrightarrow{n_2, k_2} (\alpha + k_1 + k_2,\ \ldots)$$

For the black box, starting from Beta$(2, 2)$:

$$(2, 2) \xrightarrow{41,\ 14} (16, 29) \xrightarrow{181,\ 53} (69, 157)$$

The final answer depends only on the totals $\sum k_i$ and $\sum n_i$: all 222 tries at once give the same Beta$(69, 157)$.

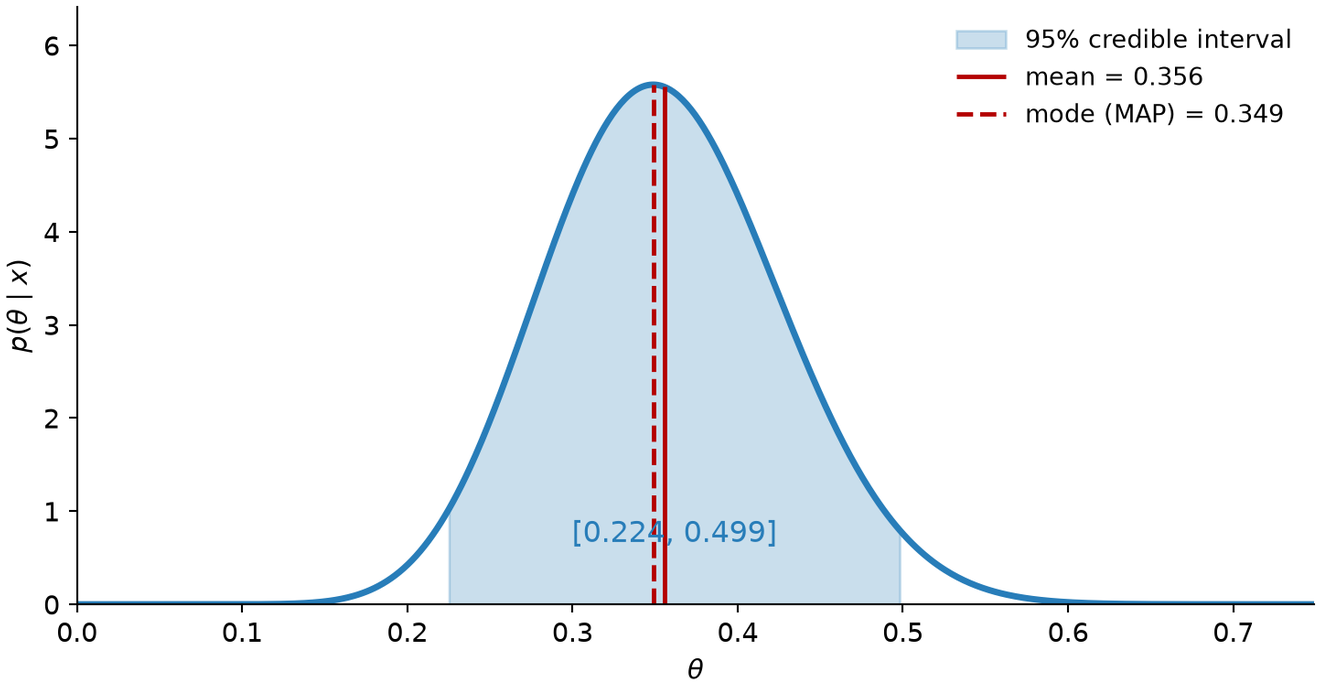
The tries are exchangeable: their probability does not change when we reorder them, so only the count of wins matters.

Streaming data

You never have to store the data, only the current posterior.

Part II: Summarising the Posterior

Summarizing The Posterior



Three summaries of one curve. The mean and the mode are not the same number.

The Posterior Mean

$$\mathbb{E}[\theta \mid x] = \frac{\alpha + k}{\alpha + \beta + n}$$

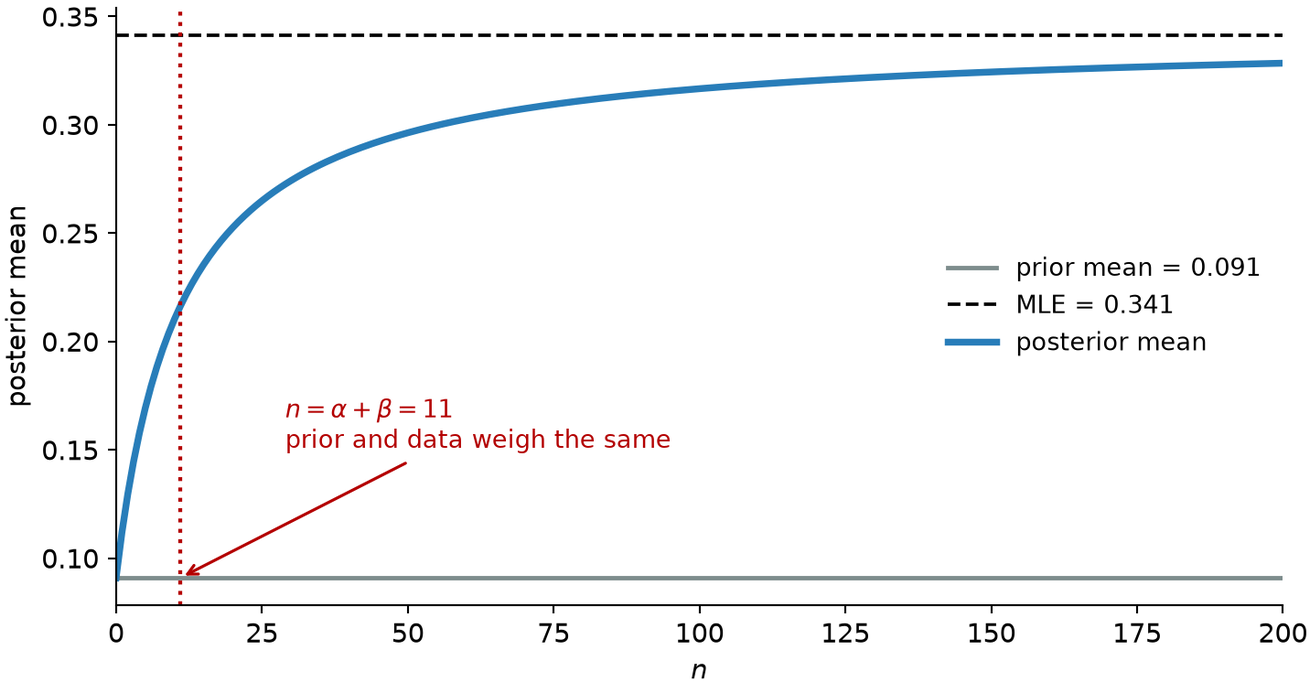
Split the numerator and it becomes something more interesting:

$$\frac{\alpha + k}{\alpha + \beta + n} = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{w_{\text{prior}}} \cdot \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha + \beta + n}}_{w_{\text{data}}} \cdot \underbrace{\frac{k}{n}}_{\text{MLE}}$$

The posterior mean is a weighted average of the prior mean and the MLE, and the two weights sum to one.

Here the prior weighs as $\alpha + \beta$ observations.

Shrinkage



With the Beta$(1, 10)$ prior, the estimate is pulled off the data towards the prior. The impact of the prior dies as n grows.

The MAP Estimate

If we insist on one number, we can take the mode of the posterior. This is the maximum a posteriori (MAP) estimate:

$$\hat{\theta}_{\text{MAP}} \;=\; \arg\max_{\theta}\; p(\theta \mid x) \;=\; \arg\max_{\theta}\; \mathcal{L}_n(\theta)\, p(\theta)$$

For the Beta$(\alpha + k,\ \beta + n - k)$ posterior:

$$\hat{\theta}_{\text{MAP}} \;=\; \frac{\alpha + k - 1}{\alpha + \beta + n - 2}$$

For our Beta$(16, 29)$: $\hat{\theta}_{\text{MAP}} = 15/43 \approx 0.349$.

MAP With a Flat Prior *Is* the MLE

Put a Beta$(1, 1)$ prior on θ. Since it is constant,it drops out of the maximization:

$$\arg\max_\theta \; \mathcal{L}_n(\theta) \underbrace{p(\theta)}_{\text{constant}} = \arg\max_\theta \; \mathcal{L}_n(\theta) = \hat{\theta}_{\text{MLE}}$$

Maximum likelihood estimation

Is Bayesian inference with a flat prior, that just keeps the location of the peak.

Credible Intervals

A 95% credible interval is any interval that holds 95% of the posterior probability:

$$P\big(\theta \in [\ell, u] \;\big|\; x\big) = 0.95$$

There are infinitely many. The usual choice is the central interval, between the 2.5% and 97.5% quantiles: for our Beta $(16, 29)$ posterior this is $[0.224,\ 0.499]$.

Another choice is the shortest such interval, here $[0.220,\ 0.495]$. Here it is nearly the same, because this posterior is almost symmetric.

Read it out loud: *given the data and the prior, there is a* 95% *probability that* θ *lies between* 0.224 *and* 0.499.

Credible $\neq$ Confidence

Credible interval

The interval is fixed; θ is random.

"Given this data and this prior, θ is in here with 0.95% probability."

Says something about **this** interval.

Confidence interval

θ is fixed; the interval is random.

"95% of intervals built this way, over repeated experiments, would cover the true θ."

Says nothing about **this** interval.

Part III: Prediction

The Question We Actually Get Asked

Nobody outside this room wants to know about θ.

They want to know: **if I run the machine ten more times, how many will I win?**

That is a question about future data, not about a parameter - and the posterior is only the first half of the answer.

Why Not Just Plug In?

The obvious move: take $\hat{\theta} = 14/41$ and use Binomial$(10, \hat{\theta})$.

But that pretends we know θ exactly. We do not - the posterior is visibly spread between 0.22 and 0.50.

Plugging in a point estimate quietly throws that uncertainty away, and the resulting prediction is too confident.

Integrate the Parameter Out

$$p(\tilde{x} \mid x) = \int_0^1 \underbrace{p(\tilde{x} \mid \theta)}_{\text{if we knew } \theta} \underbrace{p(\theta \mid x)}_{\text{how much we believe it}} d\theta$$

This is the posterior predictive distribution. Read it as a weighted average: consider every value of θ, predict under it, and weight by how credible that θ is.

The uncertainty in θ has not been discarded - it has been averaged over.

The Beta-Binomial

Doing that integral for Binomial data and a Beta(a, b) posterior, with $a = \alpha + k$ and $b = \beta + n - k$, gives

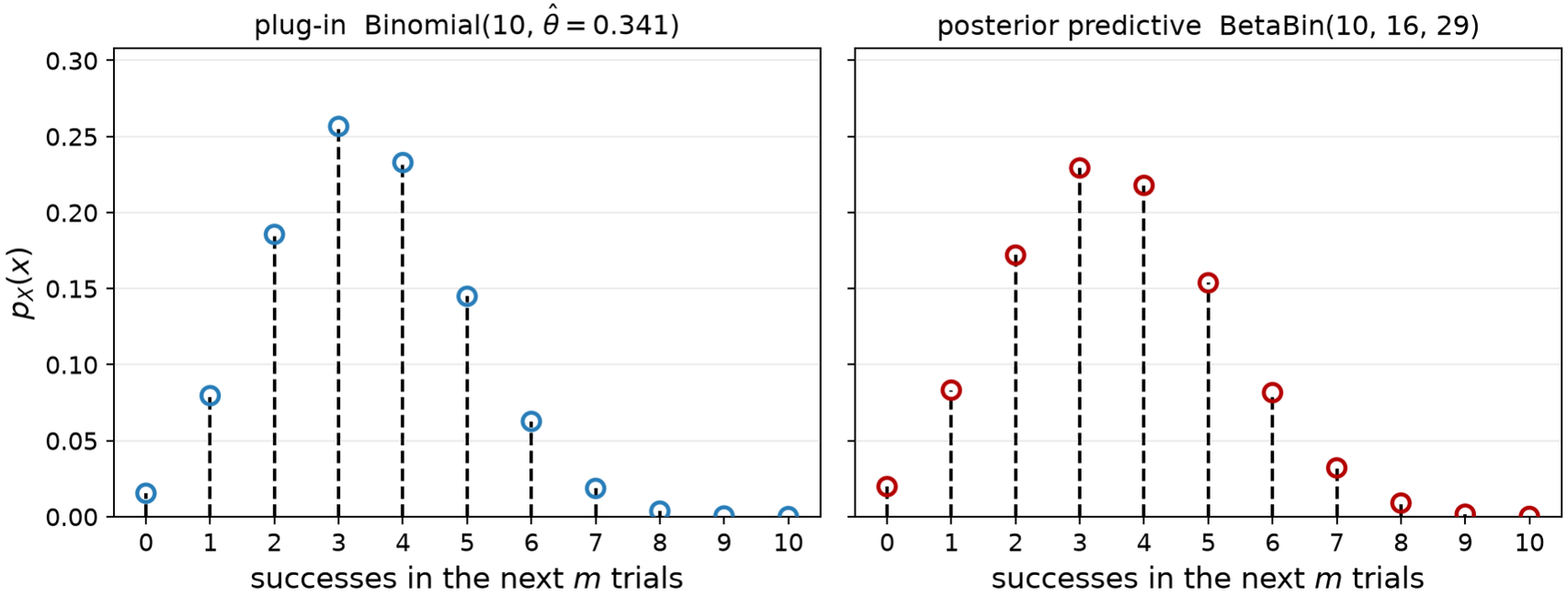
$$
\begin{aligned}
p(j \mid x) &= \int_0^1 \underbrace{\binom{m}{j} \theta^j (1-\theta)^{m-j}}_{p(j|\theta)} \underbrace{\frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)}}_{p(\theta|x)} \, d\theta \\
&= \binom{m}{j} \frac{1}{B(a,b)} \int_0^1 \theta^{a+j-1} (1-\theta)^{b+m-j-1} \, d\theta \\
&= \binom{m}{j} \frac{B(a+j,\ b+m-j)}{B(a,b)}
\end{aligned}
$$

the Beta-Binomial distribution: the number of successes in m future trials, with θ integrated out.

B is the Beta function:

$$
B(a,b) = \int_0^1 \theta^{a-1}(1-\theta)^{b-1} \, d\theta = \frac{\Gamma(a)\,\Gamma(b)}{\Gamma(a+b)}
$$

Plug-In vs Predictive



Almost the same centre. The right-hand distribution is wider, and the extra width is exactly our remaining uncertainty about θ.

The Next Single Trial

The probability of the next success (i.e. one future trial $m = 1$) is the posterior mean

$$P(\text{next trial is a win} \mid x) \;=\; \int_0^1 \theta \, p(\theta \mid x)\, d\theta \;=\; \mathbb{E}[\theta \mid x]$$

For our black box this is $16/45 \approx 0.356$ and not $14/41 \approx 0.341$.

Part IV: The Kidney Cancer Example

The Highest Death Rates

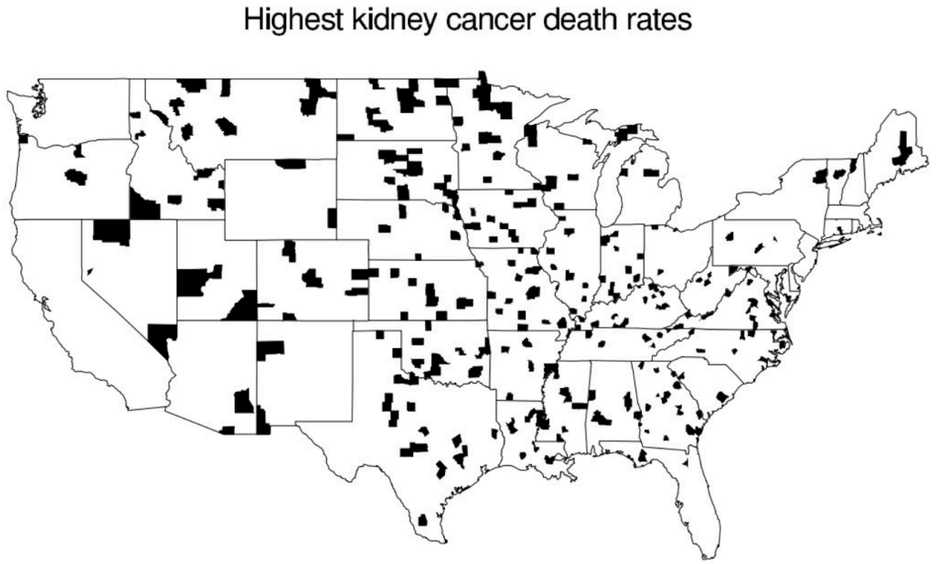


Figure 2.6 *The counties of the United States with the highest 10% age-standardized death rates for cancer of kidney/ureter for U.S. white males, 1980–1989. Why are most of the shaded counties in the middle of the country? See Section 2.7 for discussion.*

The counties in the top 10%. Mostly the rural middle of the country?

Gelman et al., *Bayesian Data Analysis*

The Lowest Death Rates

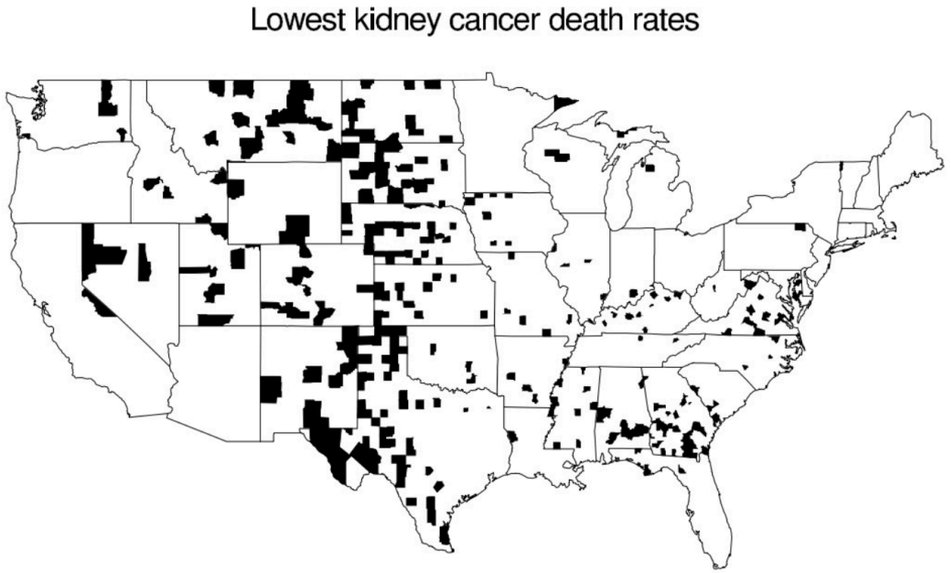
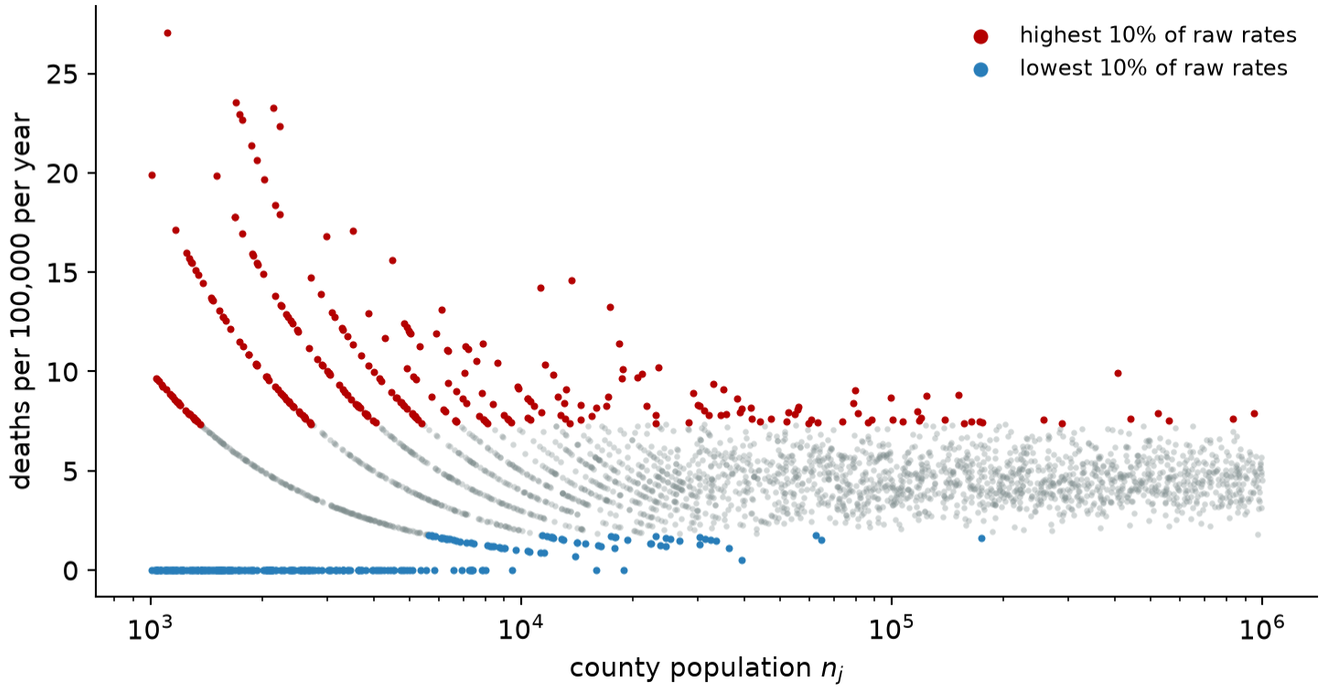


Figure 2.7 *The counties of the United States with the lowest 10% age-standardized death rates for cancer of kidney/ureter for U.S. white males, 1980–1989. Surprisingly, the pattern is somewhat similar to the map of the highest rates, shown in Figure 2.6.*

The counties in the bottom 10%. Mostly the rural middle of the country. Again!?

Gelman et al., *Bayesian Data Analysis*

Single Deaths Affect Small Counties the Most



$3{,}000$ simulated counties whose true rates all come from one distribution.

Both extremes are small counties: there, a single death more or less is the whole difference.

The Model

For county j:

- Observed, random x_j: number of kidney cancer deaths in ten years
- Observed, fixed n_j: county population
- Parameter θ_j: true death rate (deaths per person per year), one for each county

$$x_j \mid \theta_j \sim \text{Poisson}(10\, n_j\, \theta_j)$$

As a function of θ_j, the likelihood is

$$\mathcal{L}(\theta_j) \propto \theta_j^{x_j}\, e^{-10\, n_j\, \theta_j}$$

Its maximum is the raw rate, $\hat{\theta}_j = x_j/(10\, n_j)$.

A Prior for a Rate: The Gamma

A rate is positive and has no upper limit.

The Gamma distribution lives on $\theta > 0$:

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}$$

with mean α/β and standard deviation $\sqrt{\alpha}/\beta$.

Gelman's choice is Gamma$(20,\ 430{,}000)$: a mean of 4.65 deaths per $100{,}000$ per year.



Comparing the Prior to the Data

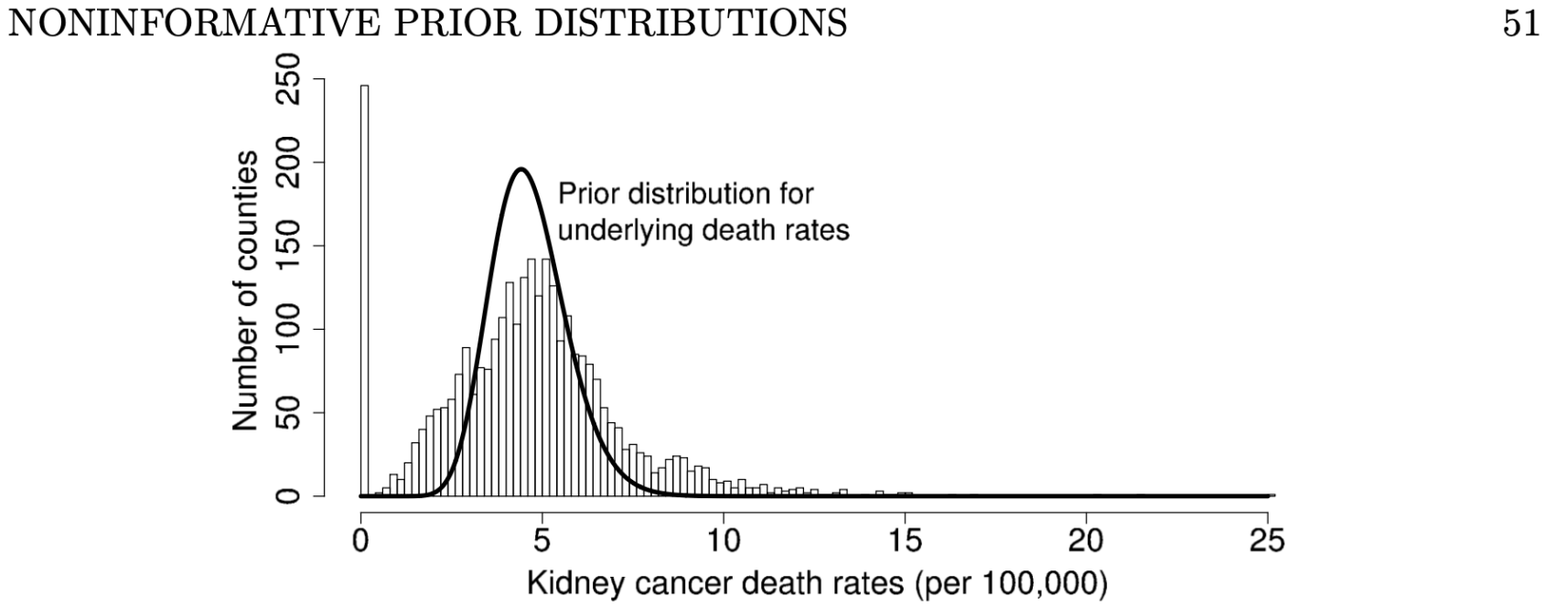


Figure 2.10 *Empirical distribution of the age-adjusted kidney cancer death rates,* $\frac{y_j}{10n_j}$*, for the 3071 counties in the U.S., along with the Gamma*$(20, 430{,}000)$ *prior distribution for the underlying cancer rates* θ_j*.*

Gelman et al., *Bayesian Data Analysis*. The figure writes y_j for our x_j.

Multiply and Collect, Again

$$p(\theta_j \mid x_j) \;\propto\; \underbrace{\theta_j^{x_j}\, e^{-10 n_j \theta_j}}_{\text{likelihood}} \cdot \underbrace{\theta_j^{\alpha-1}\, e^{-\beta\theta_j}}_{\text{prior}} \;=\; \theta_j^{(\alpha+x_j)-1}\, e^{-(\beta+10n_j)\,\theta_j}$$

The kernel of a Gamma again, so we read the answer straight off:

$$\theta_j \mid x_j \;\sim\; \mathrm{Gamma}(\alpha + x_j,\; \beta + 10\, n_j)$$

Pseudo-counts once more: the prior is worth α deaths already seen in β person-years.

Add the deaths to α, the person-years to β.

The Posterior Mean Is a Compromise, Again

$$\mathbb{E}[\theta_j \mid x_j] = \frac{\alpha + x_j}{\beta + 10\,n_j} = w_j \cdot \underbrace{\frac{\alpha}{\beta}}_{\text{prior mean}} + (1 - w_j) \cdot \underbrace{\frac{x_j}{10\,n_j}}_{\text{raw rate}}, \qquad w_j = \frac{\beta}{\beta + 10\,n_j}$$

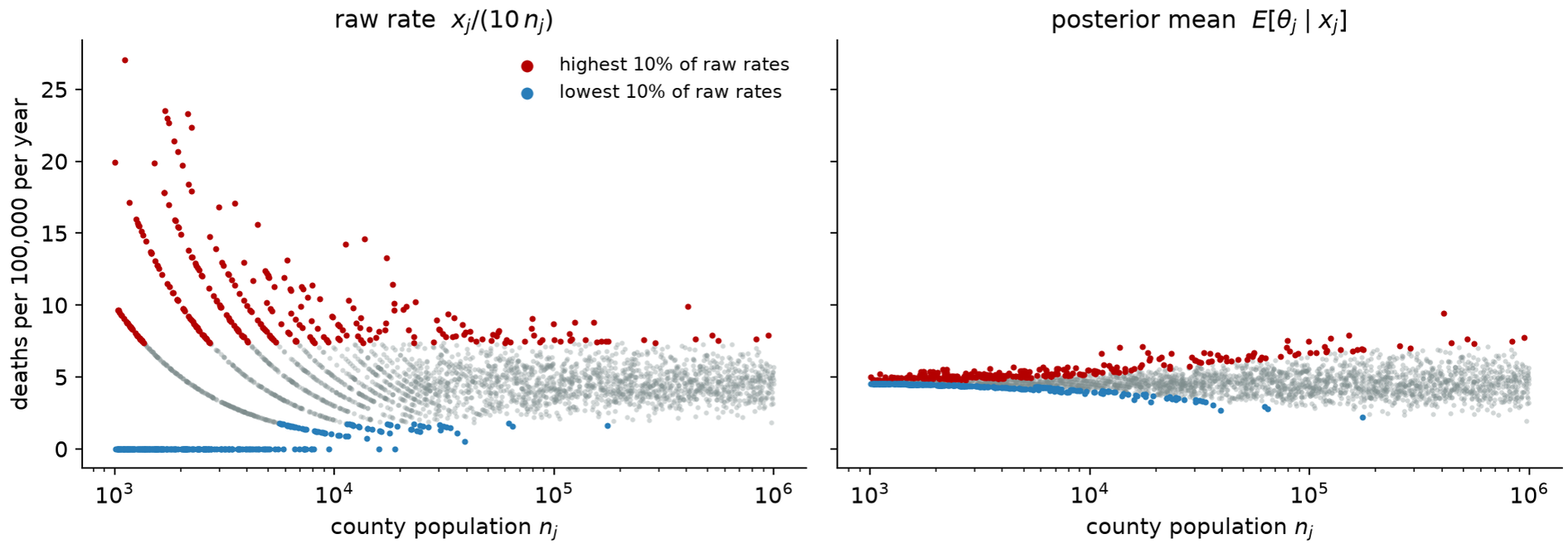
county population n_j	weight w_j on the prior
$1{,}000$	98%
$43{,}000$	50%
$1{,}000{,}000$	4%

A county of $1{,}000$ with

- No deaths has a raw rate of 0. Its posterior mean is 4.55.
- One death, 10 per $100{,}000$. Its posterior mean is 4.77.

The posterior variance is $\mathrm{Var}[\theta_j \mid x_j] = \dfrac{\alpha + x_j}{(\beta + 10\,n_j)^2}$: it shrinks as the population grows.

Shrinkage, County by County



Small counties are pulled to the prior mean; large ones speak for themselves.

This is partial pooling across counties.

The Bayesian approach allows us to estimate a large number of parameters ($3{,}000+$).

Two Conjugate Families

	Beta-Binomial	**Gamma-Poisson**
Data	k successes in n trials	x_j deaths in $10\,n_j$ person-years
Prior	Beta(α, β)	Gamma(α, β)
Posterior	Beta$(\alpha + k,\ \beta + n - k)$	Gamma$(\alpha + x_j,\ \beta + 10\,n_j)$
Update rule	successes and failures add	deaths and person-years add

In both, the posterior mean is a weighted average of the prior mean and the MLE, and with enough data the data wins.

Summary - bayesian analysis

Concept	Key formula
Bayes' rule for parameters	$p(\theta \mid x) = \mathcal{L}_n(\theta)p(\theta)/p(x)$
Proportional form	$p(\theta \mid x) \propto \mathcal{L}_n(\theta)\, p(\theta)$
Evidence	$p(x) = \int \mathcal{L}_n(\theta)p(\theta)\, d\theta$ is not a function of θ
Conjugacy	posterior in the same family as the prior
MAP	$\arg\max_\theta p(\theta \mid x)$
MAP with a flat prior	exactly the MLE
Credible interval	$P(\theta \in [\ell, u] \mid x) = 0.95$ is a statement about θ
Posterior predictive	$p(\tilde{x} \mid x) = \int p(\tilde{x} \mid \theta)p(\theta \mid x)\, d\theta$

Reading

Gelman et al., *Bayesian Data Analysis*

- Ch. 1 (p. 3–13) - probability and inference, the Bayesian setup, and the spelling corrector
- Ch. 2 (p. 29–62) - single-parameter models: the Beta-Binomial, conjugate priors, posterior summaries, the posterior predictive, and the kidney-cancer counties

Further Resources

- Will Kurt *Baysian Statistics The Fun Way*
- McElreath, *Statistical Rethinking* Book 2nd ed., youtube